

# Luke Qiao

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## EDUCATION

### Stanford University

BS/MS, Electrical Engineering (GPA: 4.1/4.0, Tau Beta Pi)

Stanford, CA

Sept 2023 – Jun 2027

## TECHNICAL SKILLS AND INTERESTS

**Analog:** Analog Circuit/IC Design, RF, Cadence Virtuoso, PCB Design, KiCAD, Altium, ADS, LTSpice, Signal Processing, DSP

**Digital:** Digital System/RTL Design, Verilog, SystemVerilog, Xilinx Vivado, Synopsys EDA Tools, FPGA, Behavioral Modeling, DV

**CS:** Python, Machine Learning, MATLAB, Linux, Computer Vision, C, C++, Java, ROS 2, Robotics, Robot Autonomy, Gazebo

## EXPERIENCES

### Apple

Jun – Sep 2026

Analog/Mixed-Signal Intern, Advanced Technologies Group

Austin, TX

- Implemented **sparse-data ML** techniques (KANs/ARD-GPs) in **SystemVerilog (SV)** for **behavioral modeling** of **PLL/SerDes blocks** (e.g. DCO, VGA, CTLE); achieved **< 1% error** with **~25x fewer** SPICE simulations, reducing Spectre compute overhead
- Built **end-to-end agentic flow** for automated ML behavioral modeling of analog blocks; cut modeling time from **weeks to hours**
- Implemented SV **pole-residue model** of **LDO PSRR** for **efficient computation** of state-transition (**~2x runtime speedup**)
- Explored **duty cycle correction (DCC)** delay-cell topologies (feedback/feedforward) to improve slew rate and reduce PSII

### Tesla

Jan – Apr 2026

Electrical Engineering Intern, Battery Electronics Team

Palo Alto, CA

- Designed/brought up an **automated benchtop test system**, validating critical battery management functions for next-gen platforms
- Led **major revision** of the benchtop tester **interface PCB** schematic & layout in **Altium**; developed **RP2040 firmware** in C++
- Validated and tuned **analog circuits** for Cybertruck, Semi, Roadster, and Cybercab battery controller boards

### Sandia National Laboratories

Jun – Dec 2025

R&D Intern / Co-op, III-V Microelectronics Group

Albuquerque, NM / Remote, CA

- Designed the first proof-of-concept rad-hard **RF amplifier PCB** with the group's custom **heterojunction bipolar transistor (HBT)** technology in LTSpice/ADS/Altium; implemented matching & distributed elements, achieving **~20 dB** gain from **1-5 GHz**
- Substantially improved HBT **noise modeling**, reducing simulation error by **60%**

### Apple

Jun – Sep 2024

CAD Engineer Intern, Silicon Engineering Group

Santa Clara, CA

- Built & deployed **optimal CPU allocation** tool in Apple's VLSI flow, reducing design verification (DV) runtime by **50%**
- Automated DV runtime data collection across 10 full chips and ~500 floorplan partitions; trained **ML models (5% error)**

### Stanford Student Space Initiative

Nov 2023 – Apr 2024

Battery Board Co-Lead

Stanford University

- Co-led **battery board thermal circuit** for SAMWISE 2U CubeSat to maintain optimal battery temperature in space; validated behavior in **LTspice**, created schematic in **KiCad** and automated heating coil trace layout with Python scripts

## SELECTED PROJECTS

**1.8 V-to-1 V, 20 mA Dual-Mode LDO in TSMC 180nm CMOS** | Cadence, Analog IC Tapeout, Layout, Design Verification

- Taped out** a **dual-mode LDO** in 10 weeks, supporting high and low power modes; designed **folded cascode & 5T OTAs**, **1V Banba BGR**, mode switching **digital logic**, and **compensation networks**; achieved **DRC/LVS clean** layout for signoff
- All **specs met** (i.e. IQ, load/line reg, phase margin, PSRR, noise) over temp/corners **post-PEX**; **bring-up** in Fall 2026

**SIMD Matrix Multiplication Accelerator** | SystemVerilog, Hardware Accelerators, Optimization, Synopsys Design Compiler

- Built and optimized a **SIMD matmul accelerator** in **SystemVerilog**, achieving **~4x throughput** via parallel MAC units, pipelining, and multiplier reuse; synthesized in **Synopsys Design Compiler**, validated with Python simulation model

**ML for Stretchable Organic IC Design** | Python, ML, IC Design, Organic Stretchable ICs, Cadence Virtuoso, MATLAB

- Built a custom **ML pipeline** to propose **stretchable OFET mechanoreceptor IC** designs from target output characteristics (**< 1% error**); created dataset from **~100k Cadence simulations**
- Trained **NN surrogate** + used **Bayesian optimization**, validated design parameters with GDA classifier (**99.4% accuracy**)

**Music Synthesizer** | Verilog, Xilinx Vivado, FPGAs, Digital System Design, Static Timing Analysis, Neural Networks

- Created a music synthesizer on a **FPGA** using **Verilog** in **Xilinx Vivado**; implemented features **purely in hardware**
- Emotion prediction w/ FNN, DSP & harmonic synthesis, ADSR envelope, ROMs, screen display via HDMI/DVI encoder

## HONORS AND AWARDS

- Tau Beta Pi Honor Society Member
- Recipient of Apple Stanford EE Cotermin Scholarship in Integrated Systems
- American Invitational Mathematics Examination Qualifier (Top 2.5%), Mathematical Association of America